

Liquid fuels demand and supply model

Manish hand-back

7 October | Baseline, fuel input tables and decision log

7 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



On customs trade, 2024 petrol and diesel sales exceed net imports by 7.7 bn litres

Corrected balance | calendar years, billion litres | imports and exports from SARS customs

Year	Petrol				Diesel			
	Sales	Imports	Exports	Sales less net imports	Sales	Imports	Exports	Sales less net imports
2019	10.77	1.48	1.17	10.47	12.91	5.59	1.74	9.06
2020	8.76	1.72	1.15	8.20	11.69	6.81	0.91	5.79
2021	9.30	4.01	1.05	6.35	12.95	9.76	0.89	4.08
2022	9.18	5.46	0.85	4.58	12.72	11.95	0.89	1.65
2023	9.04	4.48	0.99	5.55	12.91	12.87	0.89	0.93
2024	9.03	4.00	0.92	5.95	11.73	10.79	0.80	1.74
2025	—	4.45	0.80	—	—	12.25	0.75	—

Sales less net imports is what production, stock changes and gaps in sales coverage must together supply. It is not a measurement of production.

Sales: department to 2023; 2024 is FIASA's figure and is unverified; none yet for 2025. Rebuilt by one command from registered inputs; a test checks every year against the customs extract.

Sources agree on most years; five differences need a recorded choice

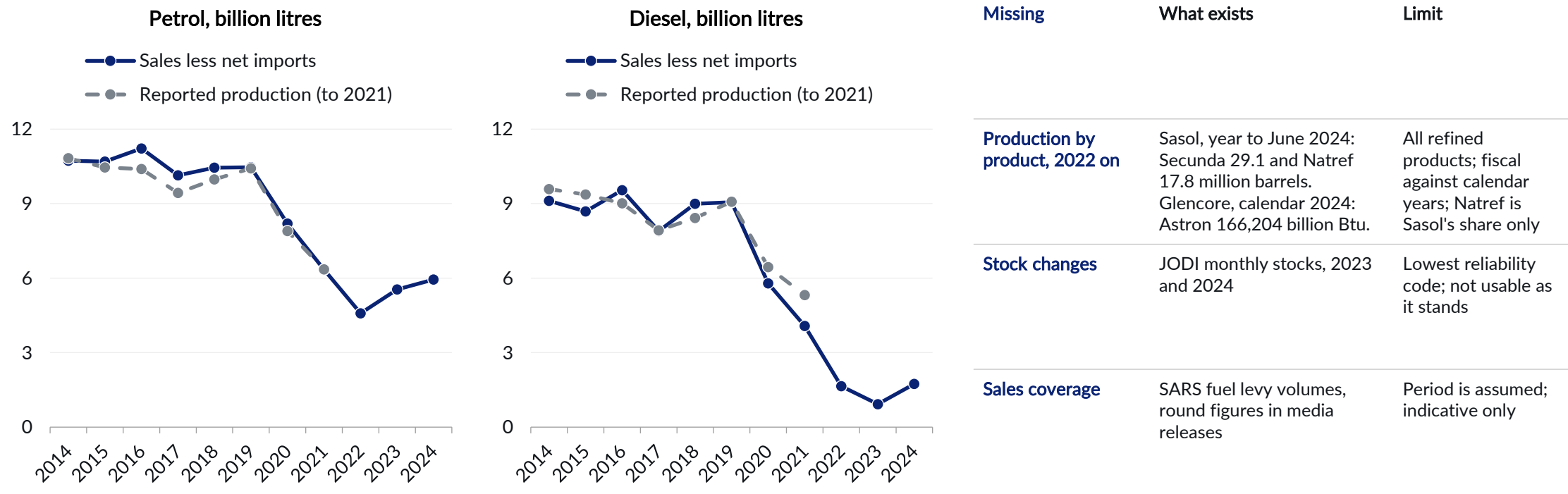
Source differences | billion litres | every source is kept in its own column of the balance file

Item	What the sources say	Used in the balance	Who decides
2024 sales	FIASA 2025 edition, p.47: petrol 9.03, diesel 11.73. FIASA 2024 edition, p.32: 8.76 and 11.81. JODI: 10.14 and 10.04 (lowest reliability code). Department: not published. Road Accident Fund levy: 24.4 bn litres for both fuels in the year to March 2025, up 0.7%, against 20.76 here.	FIASA 2025 edition, flagged unverified	Nigel: agree the 2024 sales source
2024 diesel imports	Customs 10.793. Department trade report 10.8 (rounded). FIASA 14.793, a suspected misprint.	Customs	Nigel: review
2018 and 2019 trade	2018 diesel: customs 5.12, FIASA 6.11; petrol: 1.83 and 2.19. 2019 diesel: 5.59 and 5.86. 2018 exports are also lower in customs. FIASA and the department's 2018 balance share one earlier reading; today's customs data and UN Comtrade agree with each other. No missing month or tariff line.	Customs; likely a later customs revision, not proven	Nigel: review
2022 diesel exports	Customs 0.89, FIASA 0.70.	Customs	Nigel: review
2019 and 2020 energy balances	Their import and export lines repeat the previous year's figures.	Not used for trade	Nigel: confirm

In the other years from 2014, customs and FIASA differ by 0.2 bn litres or less; the full list is in the balance note. Before 2014 customs is in kilograms or mixed units, so FIASA is used.

Production after 2021 and stock changes are missing, so the balance is not closed

Missing production and stocks | sales less net imports against production reported in the energy balance



Diesel: rail diversion is capped near 13% by rail-friendly freight; yields move up

Diesel inputs | low / medium / high for 2030 and 2035 | 11 of 42 values have a proposed replacement

Lever and unit	Baseline	2030 low / medium / high	2035 low / medium / high	Rationale and source
Road freight activity 2024 = 100	100 980 Mt road payload, 2024	95 / 115 / 135	95 / 135 / 150 was 95 / 135 / 175	Stats SA tonnes: 2.4% a year over 2014-2024, at most 3.7% over eleven years. Only tonne-km figure found: 221 bn for 2013.
Road freight moved to rail % of road freight	0 rail 161 Mt, 2024	0 / 3 / 9 was 0 / 10 / 20	0 / 7 / 13 was 0 / 20 / 35	Rail Master Plan (2026): 250 Mt target is 9%, 280 Mt market appetite 12%. Freight Roadmap (2023): rail-friendly general freight left on road is 13%.
Diesel power generation year to March 2024 = 100	100 5,143 GWh	10 / 25 / 60	5 / 15 / 50	Eskom: year to March 2026 is already 21. Agreed.
New-vehicle efficiency gain % a year	0.5 to 1.0 registered	0.5 / 0.75 / 1	0.5 / 0.75 / 1	Equals the registered endpoints, which have no source. Agreed for now.
Plant utilisation % of capacity, all fuels	67 indicative, 2024	60 / 75 / 90	60 / 75 / 90	Operators' reports: Secunda 53%, Natref 71%, Astron 84%. Secunda capacity corrected to FIASA's 150; output unchanged.
Diesel yield % of refinery output	29 JODI 2024, low reliability	25 / 30 / 40 was 20 / 25 / 30	25 / 30 / 40 was 20 / 25 / 30	JODI: 31% in 2023, 29% in 2024. Energy balances before the Durban closures: 40-42% of five fuels. Legacy 25%.
Added refining capacity thousand barrels a day	0	0 / 0 / 0	0 / 0 / 400	Conditional illustration; no investment decision sourced.

Jet: movement highs exceed every year on record; fuel per movement can also rise

Jet inputs | low / medium / high for 2030 and 2035 | 6 of 36 values have a proposed replacement

Lever and unit	Baseline	2030 low / medium / high	2035 low / medium / high	Rationale and source
Aircraft movements 2024 = 100	100 456,214 at ACSA airports	100 / 110 / 125 was 100 / 120 / 140	100 / 120 / 140 was 100 / 145 / 190	ACSA: 2019 was 112 and the 2016 peak 125. Proposed highs of 140 and 190 exceed every year on record.
Jet fuel per movement 2024 = 100	100 4,090 litres	80 / 90 / 113 was 80 / 90 / 100	65 / 80 / 113 was 65 / 80 / 100	Sales over movements: the 2013-2019 average was 4,608 litres (113). High case should allow a return to it.
Sustainable fuel share % of jet volume	0 unsourced	0 / 2 / 5	0 / 5 / 10	No South African blend requirement sourced. Agreed as a sensitivity.
Plant utilisation % of capacity, all fuels	67 indicative, 2024	60 / 75 / 90	60 / 75 / 90	Same setting as diesel and petrol; count once.
Jet yield % of capable throughput	10 legacy	5 / 10 / 15	5 / 10 / 15	Jet was 5-8% of the five fuels produced, 2017-2021 energy balances. Agreed.
Added refining capacity thousand barrels a day	0	0 / 0 / 0	0 / 0 / 400	Jet capability of any restart is not established.

Petrol: mileage and growth sit above the evidence; electric cases now follow named markets

Petrol inputs | low / medium / high for 2030 and 2035 | 22 of 42 values have a proposed replacement

Lever and unit	Baseline	2030 low / medium / high	2035 low / medium / high	Rationale and source
Passenger mileage km a vehicle a year	17,000 placeholder	12,900 / 14,500 / 17,000 was 13,600 / 17,000 / 20,400	12,900 / 14,500 / 17,000 was 13,600 / 17,000 / 20,400	Stone et al. (2018): 14,457 km for petrol cars. 2023 sales over registered petrol vehicles imply about 12,900.
Real GDP growth % a year	0.5 2024 actual	0.7 / 1.6 / 2 was 1 / 2 / 3	0.7 / 1.6 / 2 was 1 / 2 / 3	Stats SA: 0.7% a year over 2014-2024, 1.6% over 2010-2019. Treasury: 2.0% by 2028. Registered cases 1.0, 1.6.
Battery electric share % of new sales	0.2 1,088 sold, 2025	0.5 / 4 / 20 was 5 / 15 / 30	2 / 15 / 40 was 15 / 35 / 60	IEA Global EV Outlook 2026, 2025 shares: India nearly 4%, Indonesia 15%, Turkiye over 20% (from 1% in 2022), Vietnam nearly 40%.
New-vehicle efficiency gain % a year	1.0 to 1.5 registered	1 / 1.25 / 1.5	1 / 1.25 / 1.5	Equals the registered endpoints, which have no source. Agreed for now.
Plant utilisation % of capacity, all fuels	67 indicative, 2024	60 / 75 / 90	60 / 75 / 90	Same setting as diesel and jet; count once.
Petrol yield % of refinery output	54 JODI 2024, low reliability	45 / 50 / 55 was 35 / 45 / 55	45 / 50 / 55 was 35 / 45 / 55	JODI: 51% in 2023, 54% in 2024. Energy balances 2017-2021: 47-50% of five fuels. Legacy 45%. Set with diesel.
Added refining capacity thousand barrels a day	0	0 / 0 / 0	0 / 0 / 400	Conditional illustration; no investment decision sourced.

Seven levers added: price, backup generation, hybrids, rail passengers and neighbours

Added by the analyst; not in Nigel's file | low / medium / high for 2030 and 2035 | proposals for review

Lever and unit	Baseline	2030 low / medium / high	2035 low / medium / high	Rationale and source
Real diesel price 2024 = 100	100 wholesale, inland	80 / 100 / 140	80 / 100 / 140	Department and CEF prices over CPI: 76-120 in 2011-2025, 143 in October 2026. Boshoff (2012): response -0.13.
Private backup generation bn litres a year	Not measured	0 / 0 / 0.5	0 / 0 / 0.5	Diesel sales fell 1.2 bn litres in 2024 as load-shedding ended; grid turbines explain about 0.7. Indicative.
Real petrol price 2024 = 100	100 retail, inland	80 / 100 / 120	80 / 100 / 120	79-109 in 2011-2025, 121 in October 2026. Boshoff (2012): response -0.5, so the high case is about -10%.
Plug-in hybrid share % of new sales	0.5 2,810 sold, 2025	0.5 / 1.2 / 2.1	0.5 / 2 / 3.7	naamsa: 0.02% in 2022 to 0.47% in 2025. Extrapolates South Africa's own sales; not a forecast.
Conventional hybrid share % of new sales	2.1 12,818 sold, 2025	2.1 / 4.4 / 6.7	2.1 / 6.7 / 11.3	naamsa: 0.77% in 2022 to 2.15% in 2025. Extrapolates South Africa's own sales; not a forecast.
Rail passenger journeys million a year	74 2024	100 / 180 / 320	100 / 250 / 320	Stats SA: 317 in 2017, 175 in 2019, 19 in 2022, 103 in 2025. Effect in litres cannot yet be stated.
Exports to nine neighbours bn litres a year, petrol and diesel	1.48 2024	0.7 / 1.4 / 1.7	0.7 / 1.4 / 1.7	SARS: 2.66 in 2019, 1.37 in 2025; Botswana 1.05 to 0.68. Terminal throughput, not South African demand.

Sourced starting values lower 2024 diesel demand by 0.63 bn litres

Agriculture and industry | diesel, billion litres a year | proposed as provisional starting values

Sector	Earlier placeholder	On the branch	Range in the evidence	Proposed	Effect in 2024; 2035
Agriculture	0.70 no source	1.06 for 2021 energy balance, agriculture and forestry	0.90-1.09 in eight of ten years, 2012-2021; 1.89 and 1.88 in 2016 and 2017	1.06 provisional, about +/- 0.1	+0.36; +0.36
Industry	2.50 no source	1.50 for 2021 energy balance, industry; mining is 1.29 of it	1.32-1.93 over 2012-2021. Mining volume is 7% lower in 2024 than 2021, which would give 1.40	1.50 provisional, about +/- 0.3	-0.99; -1.06 (high), -1.00 (low)
Both	3.20	2.56		2.56	-0.63; -0.69 (high), -0.64 (low)

Why 2021 and not an average: it is the latest balance and one consistent year for both sectors. A 2018-2021 average gives 0.98 and 1.68; the industry average is pulled up by the years before mining output fell.

Decision for Nigel: accept both as provisional, or take the activity-adjusted 1.40 for industry. Both keep the needs-verification flag.

The totals show no double counting from the two sectors; the excess is in power generation

Overlaps and double counting | model diesel for 2024 by segment against the evidence, billion litres

Model segment	Model 2024	Evidence	Overlap or gap
Road vehicles	8.50	Balance 2021: road 6.33; commercial and public services 4.30; together 10.63	The two balance lines swap volume between years while their sum holds, so the split shows who bought the fuel, not where it was burned
Agriculture	1.06	Balance 2021: 1.06	Includes farm vehicles on public roads, which are also in the registered fleet. Size unknown
Industry	1.51	Balance 2021: 1.50; mining 1.29, construction 0.05	Mine haul trucks are unregistered, so no overlap; mines' road vehicles do overlap
Power generation	3.58	Eskom: 0.94 (year to March 2023), 0.68 (to March 2025); about 1.6 for all plants in the peak year	Not an overlap: the model is at least 2 bn litres above reported burn
Marine gas oil	0.77	No observed figure	Not established whether ships' diesel is inside inland sales
Total	15.42	Recorded sales: 12.91 (2023); 11.73 (2024, unverified)	Model is 2.5 to 3.7 above recorded sales

Road vehicles, agriculture and industry are 11.07 in the model against 13.41 of final consumption in the 2021 balance, so those three are short, not double counted. At most 2.56 could overlap with road vehicles; no source sizes it.

Lesedi: eight loading bays cap the high case near 2.4 turns a month, not 3

Vopak Lesedi, Jameson Park | proposed site assumptions | estimates are to confirm with the client

Item	Value	Basis	Source
Design capacity	140,000 m3: petrol 40,000, diesel 80,000, one 20,000 tank for either. New tanks in service since October 2025	Evidence	NERSA licence amendment, 6 August 2025; Vopak Q3 2025 release
Operational capacity	124,000 m3: 88,000 in the six original tanks; 18,000 assumed for each new tank	Evidence; new tanks estimated, to confirm with client	Vopak licence application to NERSA, 2020, p.8-9
Receipts	Pipeline only, at most 1,500 m3 an hour	Evidence	Licence application, p.9
Dispatch	At most 1,500 m3 an hour; 80% road, 20% pipeline. 8 bays at 125 m3 an hour	Evidence	Licence application, p.9; NERSA 2025
Practical road dispatch	About 300,000 m3 a month	Estimate, to confirm with client	Bays pumping half the time, 20 hours a day
Monthly turns	1 / 2 / 2.4 (low / base / high), on operational capacity	Estimate, to confirm with client	Base from Vopak's Durban target; high from road dispatch
Annual handling	1.5 / 3.0 / 3.6 million m3	Calculated	Pack today: 1.7 / 3.4 / 5.0 on gross capacity and 1 / 2 / 3 turns

Vopak's website publishes none of the operating data. Still needed from Vopak: twelve months of outbound volume by product and mode, the new tanks' operational capacity and loading hours.

Durban: fuel tanks are under half the site, so handling is far below the gross-capacity figure

Vopak Durban, Island View | proposed site assumptions | estimates are to confirm with the client

Item	Value	Basis	Source
Gross capacity	360,246 m3 for petroleum, chemicals and oleochemicals	Evidence	Vopak terminal page, checked 6 October 2026
Built for petrol and diesel	162,000 m3 in ten tanks; a floor, older tanks not counted	Evidence of a floor, to confirm with client	Vopak news release, 17 February 2017
Operational capacity, fuel	146,000 m3	Estimate, to confirm with client	90% of design, as at Lesedi
Receipts and dispatch	Vessel (five berths), pipeline, rail, truck. No rate published for any mode	Evidence; rates not found	Vopak terminal page
Pipeline to Gauteng	148,000 m3 a week for all shippers and products	Evidence	Transnet Pipelines, 24 inch trunk line
Monthly turns	1 / 2 / 3 (low / base / high), on operational fuel capacity	Target, not realised; to confirm with client	Vopak to NERSA: 'targeting 2-3 throughputs in Durban'
Annual handling, fuel	1.8 / 3.5 / 5.3 million m3	Calculated	Pack today: 4.3 / 8.6 / 13.0 on gross capacity

The high case needs 438,000 m3 a month, 68% of the whole trunk line if it all went inland by pipe. Durban-to-Lesedi volume is handled at both sites and must be counted once.

Four priorities have outputs to review; the fifth is partly done

Status against the five priorities of 7 October | ready, proposed or partial

Priority	Status	Ready to review	Not done or not available
1 National balance	Ready, with stated gaps	Balance rebuilt on customs trade with tests; source differences listed; 2018-2019 traced as far as data allows	Production by product after 2021 and stocks exist in no source found
2 Forecast levers	Proposed	20 levers with baseline, rationale and source; 39 of 120 values with a replacement; seven levers added	Some baselines are placeholders; freight is in tonnes
3 Sector baselines	Proposed	Agriculture 1.06 and industry 1.50 bn litres, with the forecast effect and an overlap table	On-road overlap not sized; power generation is 2 bn litres above reported burn
4 Tank handling	Proposed	Lesedi and Durban site assumptions; eight values marked to confirm with the client	No Vopak operating data; other operators' sites not done
5 Hand-back	Partial	This deck, the hand-back note, a 21-item decision log, and the workshop workbook with history from 2012	Pack figures not yet checked against originals; vehicle block not done

New evidence today: Road Accident Fund accounts give 24.4 bn litres of petrol and diesel levied in the year to March 2025, against 20.8 recorded as sold, and show volumes rising where FIASA shows a fall.

Decision log: 3 ready, 9 proposed, 9 unresolved

Decisions for Nigel | the nine that are unresolved, with the analyst's proposal | full log in the csv

Decision	Evidence in brief	Proposal
D01 Which 2024 national sales figure to use	Department has published nothing after 2023. FIASA 2025 edition: petrol 9.03, diesel 11.73 bn litres; FIASA 2024 edition: 8.76 and 11.81. Road Accident Fund levy shows volumes up 0.7%, not down 5%	Start the forecast from 2023; carry 2024 as unverified
D05 Treatment of the 2014 and 2018 provincial differences	Provincial sum is above the national file by 0.13-0.14 bn litres in 2014; 2018 quarter 1 district sheet is missing districts	National file stands; both records kept
D08 Secunda utilisation	Capacity corrected to 150 thousand barrels a day with utilisation re-based so output is unchanged; model gives 62 for 2024, Sasol reports 80	Raise utilisation to match reported output
D09 Reference case	No medium case is agreed; medium values are proposals	Agree after the ranges
D11 Power generation starting value	Model has 3.58 bn litres for 2024; Eskom reports 0.68-0.94 and all plants about 1.6 in the peak year	Recalibrate to reported burn at 0.31 litres per kWh
D12 Allowance for on-road use inside agriculture and mining diesel	No published source sizes it; 20% would be about 0.47 bn litres; Road Accident Fund rebates cover 1.46 bn litres across all rebated sectors	None applied; ask SARS for refund data by sector
D13 Whether ships' diesel sits inside the inland diesel total	Model carries 0.77 bn litres of marine gas oil; no observed figure	Decide scope before reconciling to sales
D17 Send the data request to Vopak	Eight values marked to confirm with client; Vopak's website publishes no operating data	Send now
D21 Marine baseline elasticities and road parameters	Carried over from 6 October; no new evidence	Unchanged

The workshop workbook now carries sourced history from 2012 and labelled provincial estimates

Demand_baseline_workshop_2026_10_08_history.xlsx | built by one command | Nigel's sheets unchanged

Sheet added	What it holds	What it shows
History	Department sales by province, customs imports and exports, refinery production, the balance and the Road Accident Fund count, 2012-2025; totals and balances are formulas	Net imports of petrol and diesel rose from 3.7 bn litres (2014) to 15.7 bn (2022). Production by product stops at 2021
History, section 6	Estimated provincial shares and volumes for 2023-2025, kept apart from observed rows	Quarter 1 2023 shares are the best of six methods tested: 2.5 share points misallocated for petrol, 5.3 for diesel
Sector history	Mining, manufacturing and agriculture diesel beside their activity measures, with litres per unit of activity	Mining swings between 11 and 17 million litres per index point; indicative 2024: mining 1.40, agriculture 0.96 bn litres
Checks	66 values on the Evidence sheet recomputed from registered inputs	All match
History sources	Publisher, link, original file, extract and refresh command for each dataset	

One change to an existing sheet: source selection for 2022 and 2023 is set to the department instead of FIASA. 2024 stays on FIASA, unverified. 2025 has no national sales figure from any source.

Review the branch and decide

manish-branch | checks pass | nothing
merged to main

7 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com

